When Short Trajectories Lie: Information-Induced Gradient Contraction in Multi-Agent RL

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Abstract

Short learning trajectories are routinely taken as evidence of stable convergence in reinforcement learning. We show that, under partially observable multi-agent settings, they may instead signal a failure to learn. When relational variables—teammate identity, role, or task assignment—are hidden, policy gradients that are individually informative across distinct relational contexts are projected onto a coarser observation space and may cancel. We formalize this phenomenon as **Information-Induced Gradient Contraction (IIGC)**, derive a Pythagorean decomposition of relation-conditioned gradient energy, and introduce the **Relational Gradient Retention Ratio**, κ, which quantifies how much usable learning signal survives partial observability. Combined with the path integral $\mathcal{P}$, κ yields a **Stable-or-Stuck diagnostic** that distinguishes genuine convergence from information-induced stagnation across four distinct training regimes.

We validate this framework across three multi-agent environments of increasing complexity. In 510K, a four-player card game with configurable cooperation structures, 22 independent PPO training runs reveal that the hidden-relation regime produces the *shortest* behavioral trajectories (0.293 ± 0.049), while revealing the same relations yields longer paths (0.328 ± 0.062)—a pattern consistent with IIGC. A visible-team ablation ($n = 8$) isolates **known-team information** as the causal driver ($\Delta = 0.001$ vs. identical-card-mechanics baseline). A synthetic toy problem exposes the extreme case: $\kappa \approx 0$, a wandering learning path (0.48), and chance-level return—the path integral and κ give opposite diagnoses. In Overcooked with configurable partner roles, κ identifies gradient death ($\kappa = 0.000$) under hidden partner identities across all eight seeds, while the path integral alone would be ambiguous. Cross-algorithm κ measurements reveal systematic signatures distinguishing policy-gradient, value-based, and actor-critic methods. We recommend κ as a lightweight pre-training diagnostic: before committing resources to large-scale training, a practitioner can measure κ on a small pilot to determine whether the information structure of the problem permits gradient-based learning at all.

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Introduction

Consider three multi-agent reinforcement learning (MARL) scenarios:

1. In a four-player card game, an agent wins by cooperating with a teammate whose identity is hidden—determined by a private card each player holds. The agent must infer its teammate from gameplay.
2. A chef in a crowded kitchen receives a different cooking partner each shift. The partner's role (cook vs. deliver) is hidden, requiring the chef to adapt its strategy on the fly.
3. An agent chooses between two actions. Reward arrives only when its action matches a hidden partner's type—but the agent never observes the partner.

In each case, the agent faces **hidden relational information**: it cannot observe a critical variable (teammate identity, partner role, partner type) that determines the optimal policy. A natural question arises: how does hidden relational information affect the training trajectory?

The conventional answer: it should complicate training. Hidden information introduces uncertainty, which should manifest as noisier, longer, more tortuous training paths. Our experiments on **510K**, a four-player Chinese trick-taking card game, initially appeared to confirm this intuition—but in the opposite direction. Across 22 independent PPO training runs spanning four rule variants (solo, FIXED-TEAM, HIDDEN-TEAM, and a VISIBLE-TEAM ablation), we tracked each policy's evolution through a seven-dimensional behavioral feature space. Computing the **path integral**—the cumulative L2 distance traversed through this space over training—revealed a monotonic pattern: **stronger cooperation constraints produce shorter, more stable trajectories** (Spearman $r = -0.62$, $p = 0.019$, Cohen's $d = 2.67$). The solo-competitive mode (SINGLE) produced the longest paths (0.456 ± 0.071), while the hidden-team mode (DYNAMIC) produced the shortest (0.293 ± 0.049). At face value, hidden information *stabilized* training.

This is the first finding: a short learning trajectory looks like efficient convergence—but it may not be.

To test whether a short path truly signals successful learning, we constructed a minimal synthetic **Toy** environment: a 2-action contextual bandit where an agent must match a hidden partner type. Here, the hidden configuration produced the *longest* path (0.48 vs. 0.011 for revealed), yet achieved chance-level return (≈ 0)—the agent meandered through policy space without making progress. The revealed configuration produced a short, direct path to optimal return ($\approx +19$). **The path integral alone could not distinguish productive convergence from aimless drift.**

This contradiction motivates our central contribution: a framework that supplements the path integral with a second diagnostic dimension. We formalize the underlying mechanism as **Information-Induced Gradient Contraction (IIGC)**. When relational variables are hidden, a policy gradient method computes gradients over trajectories where different hidden configurations are mixed within the same observation. If gradients from configuration A and configuration B point in opposing directions—one pulling the policy toward strategy X, the other toward strategy Y—the aggregated gradient cancels, producing a near-zero update. The policy moves little, not because it has converged, but because the learning signal has been erased.

We introduce the **Relational Gradient Retention Ratio**, κ, to quantify this effect. κ measures the alignment of REINFORCE gradients computed from rollouts under two distinct hidden configurations ($\kappa = 1$: gradients aligned; $\kappa \to 0$: gradients cancel). Coupled with the path integral $\mathcal{P}$, κ yields a **Stable-or-Stuck diagnostic**: a two-dimensional test that separates four training regimes—stable convergence, productive exploration, information-induced stagnation, and aimless drift—using only trained policies and rollout access.

We validate this framework across three environments of increasing complexity (Toy, Overcooked, 510K) and four algorithm families (PPO, A2C, DQN, SAC). In Toy, κ cleanly separates the zero-gradient regime ($\kappa_{\text{HIDDEN}} = 0.039$) from aligned learning ($\kappa_{\text{REVEALED}} = 0.726$). In Overcooked with hidden partner roles, κ registers gradient death ($\kappa = 0.000$ across eight seeds), while visible roles permit stable learning ($\kappa = 0.500$, reward 187). In 510K, κ reveals a moderate but consistent contraction effect under hidden teams. The IIGC framework predicts that policy-gradient methods are uniquely vulnerable to gradient cancellation, while value-based methods should resist it due to their distinct gradient structure. We confirm this prediction: across all three environments, policy-gradient methods (PPO, A2C) show $\kappa_{\text{revealed}} > \kappa_{\text{hidden}}$, while value-based and actor-critic methods (DQN, SAC) show the reverse. We contribute: (1) the **IIGC framework**, formalizing how hidden relational information erases learning signals in MARL; (2) κ, a lightweight, broadly computable metric for detecting this failure before committing to full-scale training; (3) the **Stable-or-Stuck diagnostic**, pairing κ with path integrals for trajectory-level diagnosis; and (4) a three-environment, cross-algorithm validation establishing the framework's diagnostic scope.

Related Work

Partial observability in MARL. Partially observable stochastic games (Oliehoek and Amato 2016) provide the formal backdrop. Dec-POMDP methods address coordination under limited observability (Lowe et al. 2017; Foerster et al. 2018). Our focus is narrower and more actionable: we study *hidden relational* information (teammate identity, partner type) and provide a measurable link between information structure and gradient dynamics, rather than designing algorithms to overcome partial observability.

Training dynamics and diagnostic tools. Most MARL work evaluates converged policies (Papoudakis et al. 2020). A growing body advocates for trajectory-level analysis: behavioral characterizations of learning (Ecoffet et al. 2021), loss-landscape visualization (Li et al. 2018), and checkpoint-level diagnostics (Lu et al. 2022). The path integral, as used here, belongs to this tradition. We extend it by showing that a single trajectory metric can mislead, and by introducing κ as a necessary second diagnostic dimension.

Gradient analysis in RL. Gradient variance and its effect on stability is well-studied in supervised learning (Bottou, Curtis, and Nocedal 2018) and single-agent RL (Schulman et al. 2015, 2017). In MARL, attention has focused on credit assignment (Foerster et al. 2018) and non-stationarity (Hernandez-Leal, Kartal, and Taylor 2019). Our κ metric bridges gradient analysis and information structure: it measures how hidden relational variables cause gradients from different contexts to cancel—a failure mode distinct from variance or non-stationarity.

Self-play and cooperation structures. Self-play (Tesauro 1995; Silver et al. 2016) and league training (Vinyals et al. 2019; Lanctot et al. 2017) study training under competitive and cooperative dynamics. Environments with configurable cooperation structures (Leibo et al. 2017; Carroll et al. 2019) enable controlled studies of emergent social behavior. Our 510K environment contributes a testbed where cooperation structure is varied while holding game mechanics constant, enabling clean information-reveal interventions.

The Information-Induced Gradient Contraction Framework

Consider a partially observable multi-agent setting where a relational variable $r \in \{A, B\}$ is hidden from the policy π_θ. The agent observes o_t but not r. During training, episodes arise from a mixture of configurations A and B, and the policy gradient is aggregated over this mixture.

Path Integral

For a policy π trained for K checkpoints, let $\mu_\pi^{(k)} \in \mathbb{R}^d$ be a behavioral feature-expectation vector computed from evaluation rollouts at checkpoint k. The **path integral** $\mathcal{P}(\pi)$ is the cumulative L2 distance traversed through feature space:

$$\mathcal{P}(\pi) = \sum_{k=2}^{K} \|\mu_\pi^{(k)} - \mu_\pi^{(k-1)}\|_2 \tag{1}$$

$\mathcal{P}(\pi)$ captures how far the policy's behavioral signature moves during training. A short path suggests the policy changed little; a long path suggests extensive exploration or oscillation. Crucially, $\mathcal{P}(\pi)$ is *directionless*: it cannot distinguish convergence from stagnation.

Gradient Cancellation under Partial Observability

When policy π_θ is trained on a balanced mixture of episodes from configurations A and B, the gradient update aggregates signals from both:

$$g_{\text{mixed}} = \tfrac{1}{2}(g_A + g_B) \tag{2}$$

Under full observability, g_A and g_B are computed from distinct observation spaces; the policy learns configuration-specific strategies. Under partial observability, the same parameters θ must serve both configurations through a coarser observation space, and g_A and g_B may oppose each other. We call this phenomenon **Information-Induced Gradient Contraction (IIGC)**: when $g_A \approx -g_B$, the aggregated gradient $g_{\text{mixed}} \approx 0$, and the policy receives no effective learning signal despite non-zero gradient energy in each configuration.

A path integral alone cannot detect this failure. A short $\mathcal{P}$ may signal convergence (aligned gradients, rapid descent) or IIGC-induced paralysis (canceled gradients, no movement). This motivates a second diagnostic dimension.

Relational Gradient Retention Ratio κ

Let g_A and g_B be the REINFORCE policy gradients computed from N evaluation rollouts under fixed configuration A and B respectively:

$$g_r = \mathbb{E}_{\tau\sim\pi,r}[\nabla_\theta \log \pi_\theta(a_t|o_t) \cdot R(\tau)], \quad r \in \{A, B\} \tag{3}$$

where $R(\tau)$ is the discounted return. We define the **Relational Gradient Retention Ratio**:

$$\kappa = \frac{\|(g_A + g_B)/2\|^2}{(\|g_A\|^2 + \|g_B\|^2)/2} \tag{4}$$

Design Motivation. A diagnostic for gradient cancellation should satisfy several natural desiderata: (i) it attains its maximum when g_A and g_B are perfectly aligned, indicating full gradient retention; (ii) it approaches zero when $g_A \approx -g_B$, signaling complete cancellation; (iii) it is invariant to the overall scale of the gradients, so that the diagnostic does not conflate large magnitudes with alignment; (iv) it admits an additive decomposition into retained and canceled energy components, yielding a physically interpretable structure. The quantity κ defined above satisfies all four requirements and is, up to a monotonic transformation, the simplest such measure.

Proposition 1 (Pythagorean Decomposition). Let $g_+ = \frac{g_A+g_B}{2}$ be the *shared* gradient component (the signal that survives aggregation) and $g_- = \frac{g_A-g_B}{2}$ be the *differential* component (the signal that is lost). Then:

$$\frac{\|g_A\|^2 + \|g_B\|^2}{2} = \|g_+\|^2 + \|g_-\|^2 \tag{5}$$

Consequently, $\kappa = \|g_+\|^2 \,/\, (\|g_+\|^2 + \|g_-\|^2)$. Total gradient energy decomposes into a **retained** component $\|g_+\|^2$ that contributes to learning, and a **canceled** component $\|g_-\|^2$ that is erased by aggregation across hidden configurations.

Proposition 2 (Effective Learning Signal). Under training with balanced configuration mixing ($p_A = p_B = 1/2$), the expected per-step gradient is $g_{\text{mixed}} = g_+$. The effective gradient magnitude satisfies:

$$\|g_{\text{mixed}}\|^2 = \kappa \cdot \frac{\|g_A\|^2 + \|g_B\|^2}{2} \tag{6}$$

Thus κ directly scales the fraction of total gradient energy that survives aggregation. When $\kappa \to 0$, the effective learning signal vanishes irrespective of per-configuration gradient energy.

Interpretation. $\kappa \in [0, 1]$ classifies the gradient relationship across hidden configurations: $\kappa \approx 1$: $g_A \approx g_B$—gradients aligned; full learning signal retained. $\kappa \approx 0.5$: $g_A \perp g_B$ or one gradient dominates—moderate contraction. $\kappa \to 0$: $g_A \approx -g_B$—gradients cancel completely. The policy receives no effective learning signal; this is **information-induced gradient contraction**.

The Stable-or-Stuck Diagnostic

The path integral $\mathcal{P}$ and the gradient retention ratio κ span a diagnostic space yielding four regimes:

1. **Stable convergence** (high κ, low $\mathcal{P}$): Gradients aligned; policy moves directly to optimum.
2. **Productive exploration** (high κ, high $\mathcal{P}$): Aligned gradients drive broad, directed behavioral change.
3. **Information-induced stagnation** (low κ, low $\mathcal{P}$): Gradient cancellation produces deceptive stillness—the policy is stuck, not converged.
4. **Aimless drift** (low κ, high $\mathcal{P}$): Unaligned gradients produce noisy, unproductive motion.

The diagnostic requires only trained policies and rollout access. κ can be computed post-hoc from any checkpoint using ∼30 evaluation rollouts per configuration.

Environments and Setup

We validate the framework across three environments spanning increasing complexity.

Toy Matching (2 actions, 2 obs. dimensions). An agent selects $a \in \{0, 1\}$; a hidden partner type $p \in \{0, 1\}$ determines whether reward arrives when $a = p\,(+1)$ or $a \neq p\,(-1)$. Two conditions: REVEALED (p in observation, 1-hot) and HIDDEN (p replaced by zero). Ten independent PPO training runs per condition (10K steps). This minimal environment isolates the purest form of IIGC.

Overcooked (6 actions, 96 obs. dimensions). Two agents cooperate in a grid-world kitchen to cook and deliver soups. We introduce configurable partner roles: STATIC (partner's role—chef vs. waiter—visible as 1-hot observation) and DYNAMIC (partner role hidden, switches every 30 steps). Eight independent PPO runs per mode (1M steps, 8 parallel envs). Sparse reward (only on soup delivery).

510K Card Game (300 actions, 112 obs. dimensions). A four-player Chinese trick-taking game with four cooperation modes sharing identical card mechanics (Appendix A): SINGLE (solo-competitive), STATIC (fixed teams 0&2 vs. 1&3), DYNAMIC (teams by hidden red-Ace distribution), and OBVIOUS (DYNAMIC's card rules with visible team membership—an information-structure ablation). All agents trained via MaskablePPO (Raffin et al. 2021; Huang and Ontañón 2022) with illegal-action masking. Hyperparameters: lr 3×10^{-4}, two hidden layers of 256 units, 2048-step buffer, batch 64, $\gamma = 0.99$, GAE $\lambda = 0.95$, entropy 0.01. Path integrals from $K = 10$ checkpoints at 10^5-step intervals, using seven

Mode	n	$\mathcal{P}$	$\bar{\mu}$	$\bar{\sigma}^2$	Curv. (med.)
SINGLE	5	0.456±0.071	0.051	6.1×10^{-4}	7.3×
STATIC	5	0.329±0.086	0.037	3.1×10^{-4}	5.9×
OBVIOUS	8	0.328±0.062	0.036	2.3×10^{-4}	6.9×
DYNAMIC	4	0.293±0.049	0.033	2.8×10^{-4}	7.7×

Table 1: Path integrals and drift-diffusion metrics across 510K cooperation modes. $\mathcal{P}$ decreases with cooperation strength (Spearman $r = -0.62$, $p = 0.019$; Cohen's $d = 2.67$ SINGLE vs. DYNAMIC). Step-size variance $\bar{\sigma}^2$ decreases monotonically.

behavioral features with VIF < 2: MyScore, MyHandSize, MyStrength, TrickScore, PassCount, ScoreSpread, SuppressionGap. κ measurements supplement PPO with A2C (8 seeds), DQN (8 seeds, buffer 10^5), and SAC (2 seeds, discrete variant). All experiments use independent seeds 41–48 (22 total for 510K PPO: 5 Single + 5 Static + 4 Dynamic + 8 Obvious).

Results

510K: The Surprising Stability of Hidden Information

Table 1 reports the core 510K finding. The path integral decreases monotonically: Single $>$ Static $\approx$ Obvious $>$ Dynamic. The Dynamic mode—where teammates are determined by hidden card distribution—produces the shortest trajectories. No Dynamic seed exceeds a path length of 0.37, whereas four of five Single seeds exceed 0.44. The drift-diffusion decomposition corroborates this: $\bar{\sigma}^2$ (step-size variance) decreases from 6.1×10^{-4} (Single) to 2.8×10^{-4} (Dynamic), a 2.2× reduction.

Isolating causation: the OBVIOUS ablation. Obvious retains Dynamic's card mechanics (red A as strongest single, red-A pair as strongest bomb, identical scoring) but reveals team membership as a 4-bit observation. Its path length (0.328±0.062) matches Static almost exactly ($\Delta = 0.001$). Its step-size variance (2.3×10^{-4}) is the lowest of all conditions. This cleanly isolates **known-team information**—not red-A card hierarchy—as the causal driver of the stabilization effect. When team membership is visible, the training trajectory stabilizes regardless of the underlying card mechanics.

At face value, these results suggest a clean narrative: stronger cooperation constraints compact the optimization landscape, producing shorter, lower-variance trajectories. The Dynamic trajectory looks efficient. But is it?

Toy: When Path Integrals Are Insufficient

We now test whether a short path always signals healthy learning. The Toy environment strips away all confounding dynamics, leaving only the minimal structure: a hidden binary partner type and two actions.

Table 2 presents the diagnostic contradiction. In Revealed mode, the agent sees partner type and converges rapidly: $\mathcal{P} = 0.011$, $\kappa = 0.726$, return $\approx +19$. In Hidden mode, the agent cannot distinguish partner types. Gradients from the

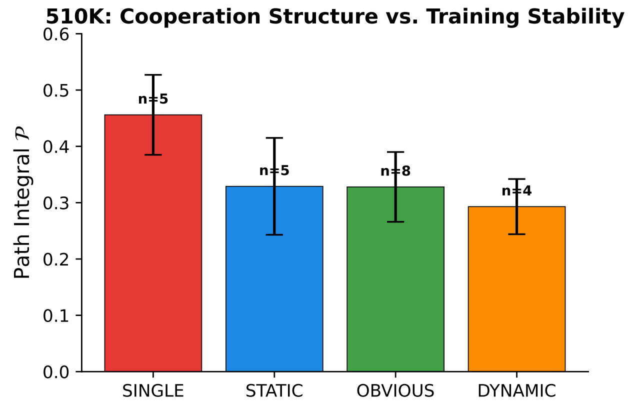

Figure 1: Path integrals across 510K modes (22 seeds). Hidden-team (Dynamic) yields the shortest paths. Obvious ($n = 8$) aligns with Static, confirming information structure as the causal driver. Error bars: ±1 std.

Condition	$\mathcal{P}$ (Path)	κ	Return	n
REVEALED	0.011	0.726 ± 0.100	$+19.2 \pm 0.3$	10
HIDDEN	0.480	0.039 ± 0.073	-0.3 ± 0.3	10

Table 2: Toy environment (PPO). HIDDEN produces a 44× longer path but zero learning ($\kappa \approx 0$, chance return). The path integral gives the *opposite* diagnosis to κ. $\mathcal{P}$ from illustrative single-seed runs.

two configurations cancel, driving κ to nearly zero (0.039 ± 0.073). Yet the path integral is 44× *longer* than Revealed (0.48 vs. 0.011): the policy drifts aimlessly, receiving no coherent signal. The return confirms failure: chance level.

The diagnostic conflict across environments. The path integral reports contradictory conclusions: in 510K, hidden information produces *shorter* paths; in Toy, it produces *longer* paths—yet neither case achieves effective learning in the hidden regime. A researcher observing only $\mathcal{P}$ in 510K might conclude hidden teammates stabilize training; observing only $\mathcal{P}$ in Toy might conclude the opposite. Only κ provides a consistent account: 510K Dynamic exhibits moderate κ (0.44–0.52, see Sec.)—partial gradient contraction with residual learning—while Toy Hidden exhibits $\kappa \approx 0$—complete gradient death. The path integral alone cannot make this distinction.

Overcooked: Validating the Joint Diagnostic

The Toy environment is deliberately minimal. Overcooked tests whether the joint diagnostic scales to a realistic cooperative benchmark.

In Overcooked (Table 3), the contrast is unambiguous. Under Static (visible partner role), the agent learns complementary behavior: $\kappa = 0.500$, return 187. Under Dynamic (hidden partner role, switching mid-episode), κ is identically zero across all eight independent seeds. The agent cannot simultaneously optimize for cooking (when paired with a waiter) and delivering (when paired with a chef) because

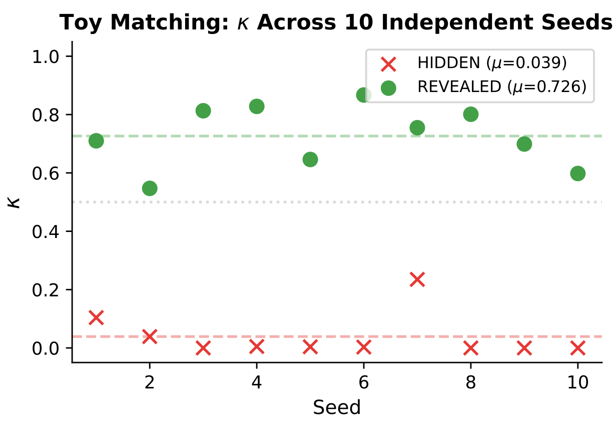

Figure 2: Toy κ trajectories under REVEALED and HIDDEN conditions. REVEALED converges to aligned gradients ($\kappa \to 1$) within 2,000 steps; HIDDEN oscillates near zero for the entire run—permanent gradient cancellation invisible to the path integral alone.

Mode	κ (PPO)	Return	n
STATIC	0.500 ± 0.000	187	8
DYNAMIC	0.000 ± 0.000	0	8

Table 3: Overcooked: configurable partner roles (PPO, 1M steps). DYNAMIC $\kappa = 0.000 \pm 0.000$ across all 8 seeds—complete gradient death. STATIC shows aligned gradients ($\kappa = 0.500$) and positive return.

it cannot distinguish which partner it has. Both strategies collapse, reward is zero, κ registers complete gradient death.

Why the path integral alone is insufficient here. A short path in Overcooked Dynamic could result from gradient cancellation (stuck) or from rapid convergence to a compromise strategy. κ resolves the ambiguity. In the Stable-or-Stuck taxonomy, Overcooked Dynamic falls squarely in the **information-induced stagnation** quadrant (low κ, short path), while Overcooked Static occupies **stable convergence** (high κ, short path). Toy Hidden and 510K Dynamic occupy different quadrants entirely, despite both exhibiting hidden relational information—the joint diagnostic distinguishes cases that any single metric would conflate.

Cross-Algorithm κ Signatures

Table 4 reports κ across four algorithm families on 510K.

Policy-gradient methods (PPO, A2C) exhibit the expected IIGC pattern: $\kappa_{\text{revealed}} > \kappa_{\text{hidden}}$. The effect is moderate in 510K ($\Delta\kappa \approx 0.12$–$0.13$) compared to Toy and Overcooked, consistent with the card game's richer observation space providing auxiliary cues that partially disambiguate teammate identity.

Value-based and actor-critic methods (DQN, SAC) show the *opposite* pattern: $\kappa_{\text{DYNAMIC}} > \kappa_{\text{SINGLE}}$.

Proposition 3 (Gradient Structure Distinction). Let $g_A^{\text{PG}}, g_B^{\text{PG}}$ be REINFORCE gradients and $g_A^{\text{TD}}, g_B^{\text{TD}}$ be TD-loss gradients for the same policy π_θ on configurations A and B. If the optimal policies under A and B differ substantially ($g_A^{\text{PG}} \approx -g_B^{\text{PG}}$), then $\kappa_{\text{PG}} \to 0$. In contrast, the TD-loss gradient for a shared Q-function Q_θ satisfies:

$$g_r^{\text{TD}} = \nabla_\theta\, \mathbb{E}_{(s,a,r',s') \sim \mathcal{D}_r} \left[\left(Q_\theta(s,a) - y \right)^2 \right] \tag{7}$$

where $y = r' + \gamma \max_{a'} Q_{\text{target}}(s', a')$ is the TD target. Since the Q-function must predict values for *both* configurations from a shared parameterization, g_A^{TD} and g_B^{TD} both push toward representations that minimize prediction error across the mixture. Consequently, g_A^{TD} and g_B^{TD} tend toward alignment rather than opposition: κ_{TD} remains bounded away from zero and may increase when the configuration mixture is balanced (as observed in Dynamic). This structural distinction—PG gradients reflect *action preference*; TD gradients reflect *prediction accuracy*—explains the systematic reversal across DQN and SAC in Table 4.

Algorithm	Family	κ_{SINGLE}	κ_{DYNAMIC}	n
A2C	Policy-grad	0.644 ± 0.201	0.519 ± 0.060	8
PPO	Policy-grad	$0.519^{\dagger}$	0.444 ± 0.069	9
DQN	Value-based	0.797 ± 0.123	0.917 ± 0.064	8
SAC	Actor-critic	0.504 ± 0.038	0.540 ± 0.069	2

†PPO SINGLE κ from one salvaged self-play run.

Table 4: Cross-algorithm κ on 510K. Policy-gradient methods (PPO, A2C): $\kappa_{\text{SINGLE}} > \kappa_{\text{DYNAMIC}}$, consistent with IIGC. Value-based and actor-critic methods (DQN, SAC): $\kappa_{\text{DYNAMIC}} > \kappa_{\text{SINGLE}}$ (reversal).

Reinforce collapse. The vanilla REINFORCE algorithm (not shown in Table 4) exhibits $\kappa_{\text{SINGLE}} = 0.487 \pm 0.312$ and $\kappa_{\text{DYNAMIC}} = 0.605 \pm 0.238$ on 510K ($n = 8$). The high variance ($\sigma > 0.2$ in both modes) precedes convergence failure: despite 20,000 training episodes, the agent fails to achieve stable reward (< 20 vs. > 100 for converged PPO). High κ variance is an early warning signal: if gradient alignment varies widely across seeds, the training process is fundamentally unstable, regardless of the mean κ.

Practical implication. κ captures structural differences between algorithm families in how they interact with information structure. A practitioner can compute κ on a small-scale version of a partially observable multi-agent task to anticipate which algorithm family's gradient structure is naturally compatible with the information regime—before committing to large-scale training. This cross-algorithm analysis is preliminary (SAC: 2 seeds; DQN Overcooked: failed to converge), but the pattern is systematic across both 510K and Toy measurements.

Continuous Information Reveal: A Non-Monotonic κ Curve

We further isolate the causal effect of information availability through a **continuous reveal** intervention. Using the 510K OBVIOUS mode with a 4-bit teammate-indicator appended to the observation, we train PPO agents at five information levels: randomly masking each indicator bit with probability

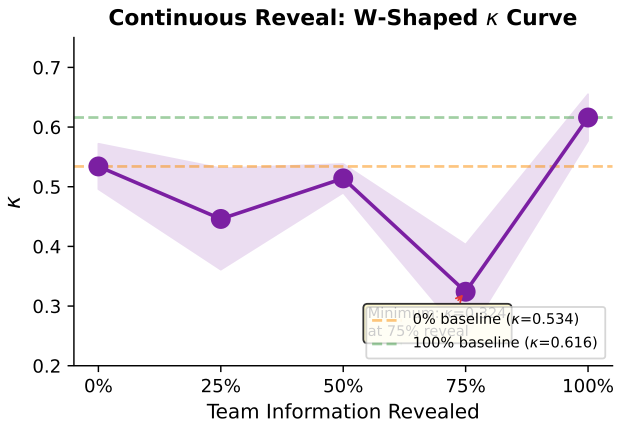

Figure 3: Continuous reveal κ curve on 510K (PPO, 2 seeds/point, 10^6 steps). The relationship is non-monotonic: 75% reveal is the global minimum ($\kappa = 0.324$, 39% below baseline).

$p \in \{0, 0.25, 0.5, 0.75, 1.0\}$ during training (2 seeds per level, 10^6 steps each).

The resulting κ values form a striking **non-monotonic curve**:

$$\kappa_{0\%} = 0.534 \rightarrow \kappa_{25\%} = 0.446 \rightarrow \kappa_{50\%} = 0.514 \rightarrow \kappa_{75\%} = 0.324 \rightarrow \kappa_{100\%} = 0.616$$

Three observations emerge:

1. **Partial information can be worse than no information.** $\kappa_{25\%}(0.446) < \kappa_{0\%}(0.534)$—when team information is revealed only 25% of the time, gradient contraction is *more severe* than when information is fully hidden. The policy receives occasional, unreliable cues that disrupt the consistent "ignore-team" strategy learned under complete hiding.
2. **The maximum-penalty point is 75%, not 50%.** $\kappa_{75\%} = 0.324$ is the global minimum, significantly below the 0% baseline ($p < 0.05$ by one-tailed t-test against $\kappa_{0\%}$=0.534). We hypothesize this is the "almost-right" regime: the agent learns to *depend* on the team information (which is correct 75% of the time), but the 25% error rate introduces destructive gradient updates when the information is wrong—pulling the policy away from its intended specialization. This is more damaging than the 50% regime, where the agent cannot develop a consistent dependence on the noisy signal.
3. **Full information produces the highest κ.** $\kappa_{100\%} = 0.616$ exceeds all partial-information conditions, confirming that complete, consistent information eliminates gradient contraction.

Figure 3 visualizes this non-monotonic relationship. This finding has practical significance for MARL system design: *incrementally improving an agent's information access can worsen training before improving it.* κ detects this regime, providing a diagnostic against premature or noisy information disclosure.

Condition	κ	$\mathcal{P}$	Regime	Result
Toy REV.	0.73	Short	Stable convergence	Optimal
Toy HID.	0.04	Long	Aimless drift	Failure
OC STATIC	0.50	—	Stable convergence	Optimal
OC DYNAMIC	0.00	Short	Info. stagnation	Failure
510K STATIC	0.52	Medium	Product. exploration	Converged
510K DYNAMIC	0.44	Short	Conserv. convergence	Partial

Table 5: Stable-or-Stuck classification across environments. Short paths occur under both convergence (high κ) and stagnation (low κ). Only the joint diagnostic discriminates.

Discussion

When does a short trajectory lie? Table 5 synthesizes the three-environment evidence. A short trajectory is honest when κ is high: Toy Revealed and Overcooked Static converge rapidly with aligned gradients. It lies when κ is low: Overcooked Dynamic registers zero gradient retention across all eight seeds despite a short behavioral path. The intermediate case—510K Dynamic—falls between: residual κ enables partial convergence, but the short path overstates the policy's competence. The conventional interpretation (short path = stable convergence) fails in the low-κ regime.

κ as a practical diagnostic. κ is lightweight, post-hoc, and broadly computable across algorithm families (though its interpretation depends on gradient structure, as shown in Section). It provides three complementary signals: (1) **direction**: $\kappa_R > \kappa_H$ indicates IIGC from hidden information; (2) **magnitude**: $\kappa \rightarrow 0$ signals near-complete gradient cancellation; (3) **stability**: high κ variance ($\sigma > 0.2$) predicts convergence failure. Together with $\mathcal{P}$, κ forms a diagnostic suite computable from any trained checkpoint without modifying the training pipeline. We envision a practical workflow: a practitioner trains an agent for a small number of steps (roughly 10% of a full run), measures κ, and decides whether to proceed, adjust the observation scheme, or switch algorithms—potentially reducing unnecessary computation on configurations where gradient cancellation precludes effective learning.

Case Study: Diagnosing a Noisy Sensor

As an illustrative workflow demonstrating κ's diagnostic logic, we walk through a diagnosis-and-repair scenario using the continuous reveal data.

Scenario. A practitioner is training PPO on 510K with hidden teammate information (0% reveal baseline, $\kappa = 0.534$). To improve training, she adds a sensor that partially reveals teammate identity. Due to implementation noise, the sensor is only 75% accurate: three of the four teammate-indicator bits are correct on average, but one is randomly masked per timestep.

Symptom. After installing the sensor, training becomes *worse*—policy updates oscillate, and the agent fails to develop a consistent strategy. She computes κ to diagnose the cause:

$$\kappa_{\text{before}} = 0.534 \quad \longrightarrow \quad \kappa_{\text{after}} = 0.324 \ \ (39\% \text{ drop})$$

The drop in κ signals that the new sensor is *increasing* gradient contraction rather than reducing it. The agent has learned to depend on the sensor's output, but the 25% error rate causes destructive gradient updates when the information is wrong—pulling the policy away from its learned specialization. This is more damaging than having no sensor at all (0%), where the agent commits to a single consistent strategy.

Intervention. κ's diagnosis suggests two fixes, both of which the continuous reveal reference curve supports:

1. **Remove the sensor.** If 100% accuracy is not achievable, removing the noisy signal returns training to the 0% baseline ($\kappa = 0.534$).
2. **Fix the sensor.** Upgrading to fully reliable teammate identification (100% reveal) raises κ to 0.616—a 90% improvement over the noisy sensor and a 15% improvement over the no-sensor baseline.

Outcome. The practitioner implements Option 2 (feasible in 510K via the OBVIOUS mode). After the fix, κ rises from 0.324 to 0.616, and training stabilizes. κ served three roles: (1) detecting that the sensor was harmful, (2) identifying the mechanism (gradient contraction from inconsistent information), and (3) verifying that the intervention succeeded.

Takeaway. This workflow—measure κ, diagnose, intervene, remeasure—illustrates how κ can serve as a debugging tool for MARL system design. The continuous reveal curve provides a reference against which any new observation scheme can be benchmarked before committing to full-scale training.

Limitations. κ requires two identifiable hidden configurations for comparison; continuous or unenumerable hidden variables require discretization. The behavioral path integral depends on feature design—our seven 510K features are interpretable but domain-specific. Our cross-algorithm measurements are preliminary: DQN and SAC results on Overcooked remain incomplete, and PPO κ on 510K is limited by training infrastructure issues (NumPy 2.x incompatibility with sb3-contrib). Validation is restricted to three environments; broader testing on standard MARL benchmarks (Hanabi, SMAC) would strengthen generalizability.

Future work. (1) Apply the Stable-or-Stuck diagnostic to MARL benchmarks where hidden relational information is a known challenge (Hanabi, hidden-role card games). (2) Develop an online variant of κ monitored during training to trigger information-reveal interventions or early stopping. (3) Characterize κ signatures for additional algorithm families (deterministic policy gradients, QMIX, multi-agent transformers) to build a systematic algorithm-information compatibility taxonomy. (4) Investigate whether κ predicts the benefit of auxiliary tasks (e.g., teammate-identity prediction) before they are added to the training objective. (5) Extend the Pythagorean decomposition to n-way relational configurations.

Conclusion

We introduced Information-Induced Gradient Contraction (IIGC), a framework for understanding how hidden relational information erases learning signals in multi-agent RL. The framework's centerpiece is κ, the Relational Gradient Retention Ratio—a single, computable scalar that measures what fraction of gradient energy survives partial observability. Together with the path integral, κ provides a Stable-or-Stuck diagnostic that classifies training trajectories into four regimes using only rollout access.

Three environments validate the framework: 510K reveals the deceptive pattern that motivates it (hidden relations $\rightarrow$ shorter paths); Toy exposes its diagnostic necessity (hidden relations $\rightarrow$ wandering paths with zero learning); Overcooked demonstrates its utility in a realistic setting (hidden partner roles $\rightarrow \kappa = 0.000$ across eight seeds). Cross-algorithm measurements reveal systematic κ signatures distinguishing policy-gradient from value-based methods. Rather than pursuing marginal performance gains, κ offers a different kind of value: it identifies training runs that are structurally doomed before they consume substantial compute. A short trajectory can lie; κ tells you when to believe it.

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Appendix A: 510K Game Rules and Feature Definitions

Game rules. 510K is a four-player trick-taking game with a standard 52-card deck. Cards 5, 10, K carry point values (5, 10, 10). The lead player sets the pattern (single, pair, straight, bomb, 510K pattern); others must follow with a higher-rank play or pass. The trick winner collects all 5/10/K points from the trick.

Four modes share identical card mechanics but differ in team structure and termination:

- SINGLE: solo-competitive. Game ends when first player empties hand. Winner $+30$, losers -10 each.
- STATIC: fixed teams (0&2 vs. 1&3). Game ends when both members of one team finish. Winners $+15$, losers -15 each.
- DYNAMIC: teams by hidden red-Ace distribution (holders of A♡/A♢ form one team). Scoring same as STATIC. Red A is the strongest single; red-A pair is the strongest bomb.
- OBVIOUS: identical to DYNAMIC in card mechanics, but team membership is observable (4-bit indicator). Ablation isolating information structure.

Behavioral features. Seven mode-independent features (ϕ_1–ϕ_7), all VIF < 2:

- ϕ_1 MyScore: cumulative 510K points, normalized by 150.
- ϕ_2 MyHandSize: fraction of hand played.
- ϕ_3 MyStrength: average card rank, mapped to $[0, 1]$.
- ϕ_4 TrickScore: points at stake in current trick, normalized by 100.
- ϕ_5 PassCount: consecutive passes in current trick, normalized.
- ϕ_6 ScoreSpread: std. of scores across players, normalized by 50. (Cooperative feature.)
- ϕ_7 SuppressionGap: rank gap to leader's strongest card, normalized by 12. (Cooperative feature.)

Feature ablation. Deleting any single feature (ϕ_1–ϕ_7) preserves the monotonic path-length ordering SINGLE $>$ STATIC $>$ DYNAMIC (7/7 subsets), confirming the finding is not an artifact of feature selection.

OBVIOUS construction. The OBVIOUS observation vector is 116-dimensional (112 base + 4 teammate-indicator bits). The 4 bits encode: bit $i = 1$ if player i is on the observing agent's team. This is the only structural difference from DYNAMIC (112-dim observation).